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Preparation of *Ocimum tenuiflorum* essential oil loaded bioactive nano-formulation

Rafia Rehman[a,b], Aqsa Younas[b], Afsar Bano[c], Rabeea Muzaffar[d], Marghoob Ahmad[e], Esha Zaman[d], Fatima Tuz Zahra[d], Imrana Mumtaz[d]*

[a]Section of Phytochemistry and Natural Products, Department of Biological Sciences, National University of Medical sciences, Rawalpindi, 46000, Punjab, Pakistan

[b]Department of Chemistry, Faculty of Sciences, University of Okara, Okara 56130, Pakistan

[c]Department of Physics, Syed Babar Ali School of Science and Engineering, Lahore University of Management Sciences, LUMS 54792, Lahore, Pakistan

[d]Department of Biological Sciences, National University of Medical Sciences, Rawalpindi, 46000, Punjab, Pakistan

[e]School of Chemical and Materials Engineering, National University of Science and Technology, Islamabad, 44000, Islamabad, Pakistan

* Email**:** rafia.rehman@numspak.edu.pk

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Abstract: Essential oil based nano-formulations are need of the time as the efficacy and stability of such formulations have valuable applications in the field of medicine and cosmetic research. Hence, in present study, the essential oil from *Ocimum tenuiflorum* plant was extracted by hydro-distillation method. The compositional ingredients used to formulate nanoemulsion by low energy spontaneous method. The physical characteristics of emulsion were evaluated by analyzing pH, viscosity, conductivity and surface tension. The droplet size and zeta potential of the nanoemulsion was analyzed by using a Zetasizer. The essential oil was then added in the prepared nanoemulsion in different concentrations, resulting in the formation of 5 samples (F_1 to F_5). The samples showed significant antioxidant activity as tested by DPPH method. Mosquitoes' repellency test of these essential oil loaded nanoemulsions showed that the activity increase with increasing e. oil concentration with 100 % repellency for F5. The results of present study suggested that the prepared formulation can be used as an important ingredient to formulate mosquito repellent products.